<u>Segment ineq.</u> $\mathrm{Ric}_{M^n} \geq -(n-1)$

$$A_1, A_2 \subset B_r(p) \quad , r \leq R \quad , g \geq 0$$

$$\int_{A_1 \times A_2} \mathcal{F}_g(x_1, x_2) \leq c(n,R)\, r \left(\mathrm{Vol}(A_1) + \mathrm{Vol}(A_2)\right) \int_{B_{2R}(p)} g$$

<u>E.g.</u> Poincaré inequality:

Suppose $\mathrm{Ric}_{M^n} \geq -(n-1)$, $\partial B_{3r}(p) \neq \emptyset$

$f : B_r(p) \longrightarrow \mathbb{R}$ (extend f by zero on $B_{3r}(p) \setminus B_r(p)$)

$f = 0$ on $\partial B_r(p)$

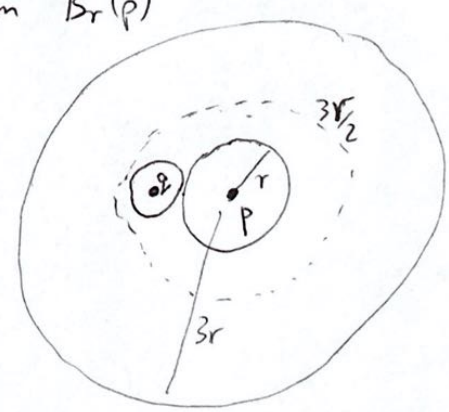
Let $g = |\nabla f^2|$

Then $|f(x_1)^2 - f(x_2)^2| \leq \mathcal{F}_g(x_1, x_2)$

$A_1 = B_r(p)$

$A_2 = B_{r/4}(q)$ disjoint from $B_r(p)$

$A_1, A_2 \subset B_{3r/2}(p)$



Segment ineq $\Rightarrow$

$$\mathrm{Vol}(A_2)\int_{\substack{A_1\\ \shortparallel \\ B_r(p)}} |f(x_1)|^2 \le \int_{A_1\times A_2} \mathcal{F}_g(x_1,x_2) \le c'(n,r)\left(\mathrm{Vol}(A_1)+\mathrm{Vol}(A_2)\right) \cdot \int_{B_{3r}(p)} |\nabla f^2|$$

$$\le 2\left(\int_{B_r(p)} |f|^2\right)^{1/2}\left(\int_{B_r(p)} |\nabla f|^2\right)^{1/2}$$

$$\Longrightarrow \left(\int_{B_r(p)} |f|^2\right)^{1/2} \le 2c'(n,r)\underbrace{\left(\frac{\mathrm{Vol}(B_r(p))}{\mathrm{Vol}(B_{r/4}(q))}+1\right)}\left(\int_{B_r(p)} |\nabla f|^2\right)^{1/2}$$

Volume comparison $\Longrightarrow$ $\dfrac{\mathrm{Vol}(B_r(p))}{\mathrm{Vol}(B_{r/4}(q))} \le c''(n,r)$

So
$$\int_{B_r(p)} |f|^2 \le C(n,r)\int_{B_r(p)} |\nabla f|^2$$

<u>Iterated segment inequality</u>

Let $A_1, A_2 = B_{R/4}(p)$, fix $\underline{x} \in B_{R/4}(p)$, $g \ge 0$.

$$\mathcal{F}_g(\underline{x},\cdot) : B_{R/2}(p) \longrightarrow \mathbb{R}$$

Can apply segment ineq. to $\mathcal{F}_g(\underline{x},\cdot)$ in place of g:

$$\int_{B_{R/4}(p)\times B_{R/4}(p)} \mathcal{F}_{\mathcal{F}_g(\underline{x},\cdot)}(y,z) \le c(n,R)\,\mathrm{Vol}(B_{R/4}(p)) \cdot \int_{B_{R/2}(p)} \mathcal{F}_g(\underline{x},\cdot)$$

Integrate this over $\underline{x} \in B_{R/4}(p)$:

$$\int_{B_{R/4}(p)} \left(\int_{B_{R/4}(p) \times B_{R/4}(p)} \mathcal{F}_{\mathcal{F}_g(\underline{x},\cdot)}(y,z) \right) d\underline{x} \leq c'(n,R)\, \mathrm{Vol}(B_{R/4}(p)) \cdot \mathrm{Vol}(B_{R/2}(p)) \cdot \int_{B_R(p)} g$$

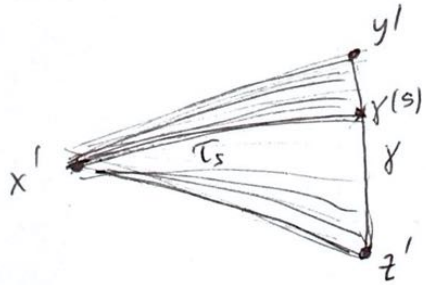
<u>Way to use this:</u> if $\int_{B_R(p)} g$ is very small

$\Longrightarrow$ $\forall\, \underline{x} \in B_{R/4}(p)$ $\exists\, x'$ near $\underline{x}$ s.t.

$$\int_{B_{R/4}(p) \times B_{R/4}(p)} \mathcal{F}_{\mathcal{F}_g(x',\cdot)}(y,z)\, dy\, dz \text{ is v. small}$$

$\Longrightarrow$ $\forall\, y, z \in B_{R/4}(p)$ $\exists\, y', z'$ near y, z s.t.

$\mathcal{F}_{\mathcal{F}_g(x',\cdot)}(y',z')$ is v. small.



$$\mathcal{F}_{\mathcal{F}_g(x',\cdot)}(y',x') = \int_0^{d(y',z')} \mathcal{F}_g(x', \gamma(s))\, ds$$

$$= \int_0^{d(y',z')} \int_0^{\ell(s)} g(\tau_s(t))\, dt$$

Comparison functions

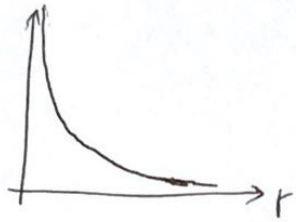
On $\underline{M}^n_H$ there are explicit functions

$\underline{G}(r)$, $\underline{U}(r)$ s.t.

$\underline{\Delta}\,\underline{G} = 0 \quad (r \neq 0)$

$\underline{G}(0) = \infty$

$\underline{G}' < 0$



$\underline{G}(d(p,\cdot))$ is Green's function with pole at p.

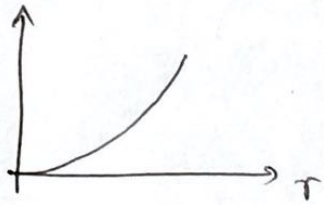
On $\mathbb{R}^n$ $(n>2)$ $\quad \underline{G} = \frac{1}{c_n} r^{2-n}$

(by Laplacian comparison if $Ric_{M^n} \geq (n-1)H$, then $\Delta\,\underline{G}(d(p,\cdot)) \geq 0$)

$\underline{\Delta}\,\underline{U} = 1$

$\underline{U}(0) = 0$

$\underline{U}' > 0$



On $\mathbb{R}^n$ $\quad \underline{U} = \frac{r^2}{2n}$

(if $Ric_{M^n} \geq (n-1)H$ then $\Delta\,\underline{U}(d(p,\cdot)) \leq 1$)

<u>Example application</u> Let $Ric_{M^n} \geq (n-1)H$

$\Omega \subset M^n$ bounded

$f : \bar{\Omega} \longrightarrow \mathbb{R}$ s.t. $\Delta f \geq \delta > 0$

$\rho_x = d(x, \cdot)$

$\Delta \underline{u}(\rho_x) \leq \Delta \underline{u} = 1$

$\Longrightarrow \Delta(f - \delta \underline{u}(\rho_x)) > 0$

maximum principle $\Longrightarrow f(x) \leq \sup_{\partial\Omega} (f - \delta \underline{u}(\rho_x))$

(quantitative max principle)

<u>Gradient estimate</u>

<u>Thm</u> (special case of Cheng-Yau) Let $Ric_{M^n} \geq (n-1)H$

$u : B_{R_2}(p) \longrightarrow \mathbb{R}$

s.t. $\Delta u = \lambda$ $\lambda =$ constant.

$\Longrightarrow$ for $R_1 < R_2$

$$\sup_{B_{R_1}(p)} |\nabla u| \leq C(n, H, R_1, R_2, \lambda) \sup_{B_{R_2}(p)} |u|$$

<u>Proof</u> same as on HW 1 :

find $\eta = \begin{cases} 1 & \text{on } B_{R_1}(p) \\ 0 & \text{near } \partial B_{R_2}(p) \end{cases}$

s.t. $\eta = \eta(\rho_p)$, $\eta' \leq 0$

$\Longrightarrow \Delta\eta \geq -c(H, R_1, R_2)$

$\exists\, D = D(\lambda, R_1, R_2, n, H)$ s.t.

$$\Delta(\eta^2 |\nabla u|^2 + D|u|^2) \geq 0$$

$\square$

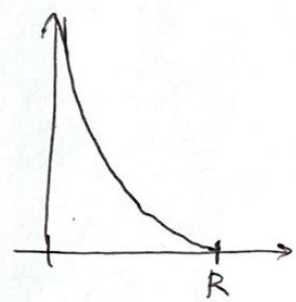
Good cutoff functions (Cheeger-Colding)

Let $\mathrm{Ric}_{M^n} \geq -(n-1)$, $0 < R_1 < R_2$. $\exists\, \phi : M^n \to [0,1]$ such that

$$\phi = 1 \quad \text{on } B_{R_1}(p)$$
$$\phi = 0 \quad \text{on } M \setminus B_{R_2}(p)$$
$$|\nabla \phi|,\ |\Delta \phi| \leq C(n, R_1, R_2)$$

Proof Recall $\underline{G}, \underline{U}$. Let $\underline{G}_R = \underline{G} - \underline{G}(R)$
$\underline{U}_R = \underline{U} - \underline{U}(R)$

for suitable $c_R > 0$

$$\underline{L}_R = c_R \underline{G}_R + \underline{U}_R$$

satisfies
$$\Delta \underline{L}_R = 1$$
$$\underline{L}_R' \leq 0$$
$$\underline{L}_R(R) = 0$$